Class: BE-Computer; Semester: VIII
Subject: Applied Data Science Lab
Experiment No.: 04

AIM- Implement and explore performance evaluation metrics for Data Models (Supervised)

Theory -

Classification -

1)Evaluate models using the following metrics

Metric	Formula	Evaluation Focus
Accuracy	$ACC = \frac{TP+TN}{TP+TN+FP+FN}$	Overall effectiveness of a classifier
Error rate	$ERR = \frac{FP+FN}{TP+TN+FP+FN}$	Classification error
Precision	$PRC = \frac{TP}{TP+FP}$	Class agreement of the data labels with the positive labels given by the classifier
Sensitivity Specificity	$SNS = \frac{TP}{TP+FN}$ $SPC = \frac{TN}{TN+FP}$	Effectiveness of a classifier to identify positive labels How effectively a classifier identifies negative labels
ROC	$ROC = \frac{\sqrt{SNS^2+SPC^2}}{\sqrt{2}}$	Combined metric based on the Receiver Operating Characteristic (ROC) space [53]
F_1 score	$F_1 = 2\frac{PRC \cdot SNS}{PRC+SNS}$	Combination of precision (PRC) and sensitivity (SNS) in a single metric
Geometric Mean	$GM = \sqrt{SNS \cdot SPC}$	Combination of sensitivity (SNS) and specificity (SPC) in a single metric

Recall=sensitivity

False positive rate (α) = type I error = 1 − specificity = FP / (FP + TN)

False negative rate (β) = type II error = 1 − sensitivity = FN / (TP + FN)

Power = sensitivity = 1 − β

2) Plot ROC curve

Regression 1)Evaluate models using the following metrics

Evaluation Metrics	Formula	Layman's explanation
Pearson correlation coefficient (r)	$r = \frac{\sum(x-\overline{x})(y-\overline{y})}{\sqrt{\sum(x-\overline{x})^2}\sqrt{(y-\overline{y})^2}}$ Where, $\overline{x}$ - mean of X variable $\overline{y}$ - mean of Y variable	Measure of the strength of the association between the two variables. In terms of predictive accuracy of the model, i.e., the Pearson correlation coefficient (r) between the predicted values and the observed values.
Coefficient of determination (R2)	$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$	Equals the fraction of variance of outcome/response explained by the model. Also, mostly referred to as goodness of fit.
Mean Squared Error (MSE)	$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	Most commonly used loss function for regression. It tells you how close a regression line is to a set of data points.
Root Mean Squared Error (RMSE)	$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$	Estimate to measure the error of the numerical predictive model. In short, measure the error rate of a regression model. Also, commonly referred as Root Mean Square Deviation (RMSD).
Root Mean Squared Relative Error (RMSRE)	$\text{RMSRE} = \sqrt{\frac{\sum_{i=1}^{s}\left(\frac{y_i - \hat{y}_i}{y_i}\right)^2}{s}}$	Here the error will be considered relatively which means that each residual is scaled against actual value or normalized by that. A residual is the vertical distance between a data point and the regression line. Each data point has one residual. Lower the value better we can establish the quality of model prediction.
Mean Absolute Error (MAE)	$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i \rvert$	It is very similar to MSE in the sense that here also we take the difference between the predicted and the actual value, and divide it with the number of testing sample values, however, instead of squaring this difference, we take an absolute value of it.
Mean Absolute Percentage Error (MAPE)	$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left\lvert\frac{y_i - \hat{y}_i}{y_i}\right\rvert$	Also, used as a loss function for regression problems. It measures the accuracy as a percentage. As MAPE is the measure of error, higher values denote higher error, while lower values represent good accuracy. Also, commonly referred as Mean Absolute Percentage Deviation (MAPD).

where y_i is the observed value, $\hat{y}_i$ is the predicted value, and $\overline{y}$ is the mean value of the observed values, **s/n** denotes the sample size of the testing dataset.